National University
of Computer & Emerging Sciences

EE Department
(Karachi Campus)
Complex Engineering Problem

Course Code: CS2011 | Fundamentals of Databases Spring 2026 | Date: 20/03/2026

Student Name: ______________________ Section:______ Student Roll No :_______

INSTRUCTIONS:

1. This CEP is a group assessment. A group is of 3 or 2 students.
2. There will be demonstration and report session along with the submission.
3. The submission date of CEP is Friday, 1st May, 2026.

1. Problem Description

With the rapid advancement of electric vehicles and large-scale energy storage systems, Battery Management Systems (BMS) have become increasingly reliant on data-driven intelligence to ensure safety, reliability, and optimal performance. Modern battery systems continuously generate large volumes of time-series data through embedded sensors, including voltage, current, temperature, and derived parameters such as State of Charge (SOC) and State of Health (SOH).

Traditional data handling approaches are insufficient to efficiently manage, analyze, and extract insights from such high-frequency, multi-dimensional datasets. Additionally, modern BMS must support multiple battery chemistries (e.g., NMC, LFP, LTO), monitor charging and discharging cycles, detect faults (such as over-voltage and over-temperature), and track degradation patterns over time.

This problem involves the design and implementation of a **relational database system** for a smart BMS that addresses challenges of scalability, data integrity, real-time monitoring, and analytical querying. The system must support structured storage of battery data, enable efficient querying for performance evaluation, and facilitate intelligent decision-making for battery safety and optimization.

The solution requires the development of an **Enhanced Entity-Relationship (EER) model**, its transformation into a normalized relational schema, and implementation of advanced SQL-based analytics. Additionally, the system must incorporate database programming constructs such as triggers for automatic fault detection and stored procedures for recommending charging profiles based on battery condition.

The complexity of the problem lies in handling time-series data efficiently, modeling hierarchical battery types using specialization, ensuring normalization while maintaining performance, and enabling advanced analytical queries for degradation analysis and fast-charging optimization.

2. Targeted Attributes

The tested attributes of the CEP are marked in the given table.

Attribute	Code	Description	Tested?
Depth of knowledge	WP1	Cannot be solved without the in-depth engineering expertise which involves design and development of solution, investigation, modern tool usage, societal impact, and ethics	✓
Range of Conflicting requirements	WP2	Involve a variety of technological, engineering, and other topics that are broad or contradictory.	✓
In-depth analysis requirement	WP3	Have no apparent solution and necessitate abstract thought and research to devise appropriate models.	✓
Familiarity of issues	WP4	Involve problems that aren't usually experienced.	✓
The extent of applicable codes	WP5	They are issues that are not covered by technical engineering guidelines and codes of practice.	
Stakeholder involvement	WP6	Involve a variety of stakeholders with a wide range of needs.	
Interdependence	WP7	They are any high-level issues including a large number of parts or sub-problems.	

3. Justification

In this problem, four different attributes of the CEP are involved. The justification of each attribute's involvement is given below.

Depth of Knowledge Required:

The problem demands expertise in database design (EER modeling, relational schema), normalization (up to 3NF), and advanced SQL programming including triggers and stored procedures. Additionally, understanding battery systems and their parameters is required.

Range of Conflicting Requirements:

The design must balance normalized structure (to reduce redundancy) with query performance for large time-series datasets. Trade-offs arise between efficient storage and fast analytical queries.

In-depth Analysis Requirement:
Students must analyze battery degradation patterns, detect abnormal SOH drops, evaluate fast-charging impacts, and compare battery chemistries under similar conditions.

Familiarity of Issues:

This problem integrates database systems with energy storage applications, requiring interdisciplinary understanding of electrical systems, data analytics, and database system design.

4. Plan of Work

This CEP will be completed in five different phases, the description of each phase along with its due date and marks are given in the table below.

Phase No.	Description	Due Date	Total Marks
1	Requirement Analysis & Data Collection • *Dataset selection (real or synthetic), system requirements*	10 April 2026	10
2	EER Model Design • *Specialization (battery types), entities, relationships*	16 April 2026	20
3	Relational Schema & Normalization • *Schema design, PK/FK, 3NF justification*	22 April 2026	20
4	SQL-Based Data Analysis • *Queries for degradation, faults, and performance*	28 April 2026	20
5	Database Programming • *Trigger (fault detection), stored procedure (charging recommendation)* Final Report Submission • *Prepare and submit the final report/user manual according to given template.*	1st May 2026	30

5. Report Format

Prepare your final report according to the given guidelines.

5.1. Contents of Report

The following items must be in your report.

1. **Title Page**
2. **Table of contents**
3. **Abstract** (write a structured abstract)
4. **Introduction** (introduce your project briefly in this section)
5. **Nomenclature** (define the terms and abbreviations used in report)
6. **Theory and Calculations** (describe the theory involved in your project design, and also mention the working principles along with all the calculations performed)
7. **Test Items** (define those items with their brief working which are used in your project for testing, or the items that need to be tested before inserting them in the project)
8. **Equipment and Apparatus** (list of all items used)
9. **Cost Analysis** (cost of each component and total cost)
10. **Procedure** (the complete method you followed for project design. Each and every step from planning to final touch. You may insert block and circuit diagrams in this section)
11. **Result** (The final result of your attempt)
12. **Conclusion** (Conclude the whole project by comparing the ideal one with your practical results)
13. **References** (Provide the reference list in IEEE style)
14. **Appendices** (You can include datasheets and any other helping reference in this section)

5.2. Length and Formatting

- There is no specific limit on the length of the report, but with the above-mentioned contents your report will easily cover a minimum of 20 pages (excluding title page and appendices).
- Follow the same font throughout of report and same heading, sub heading size/style.
- The text must be justified on both sides of the paper.
- Do not use smiley and informal words.
- Do not increase the line spacing and don't insert blank lines to fill up more and more pages.
- Insert the page number at the bottom centre.

5.3. AI Content, Plagiarism, and Similarity

- Turnitin similarity index must be less than 19% with less than 5% from a single source.
- AI percentage must be less than 20%.
- Similarity between the report of peers must be 0%.

6. Assessment Rubrics

Every deliverable will be assessed separately; however, generic rubrics are given below for each student.

Rubric for Complex Engineering Problem

Problem solving is the process of designing, evaluating, and implementing a strategy to answer an open-ended question or achieve a desired goal.

S. No.	Quality → Criteria (as per PEC Manual) →	No/Limited Proficiency (0-1point)	Some Proficiency (2 points)	Proficiency (3 points)	High Proficiency (4-5 points)	N/A	Rating (Each out of 5)
1	Depth of knowledge required: Require research-based knowledge.	Demonstrates poor research-based knowledge in regard to the implemented solution	Demonstrates inadequate research-based knowledge, the implemented solution having very little depth regarding research	Demonstrates a high level of research-based knowledge with the implemented solution, the level of research done is adequate	Demonstrates remarkable research-based knowledge, the implemented solution uses research-based knowledge in novel ways to approach the problem.		X1
2	Depth of analysis required: Have no obvious solution, require abstract thinking, originality in analysis to formulate suitable model.	Analysis of solution is very basic and shows no level of abstract thinking, the solution also lacks any form of originality regarding the formulation of a suitable model.	Analysis of solution is brief and shows limited levels of abstract thinking, but the solution lacks originality to form a suitable model.	Analysis of solution is adequate and shows higher level of abstract thinking and shows originality in implementation of a suitable model.	Analysis of solution is deep and effective containing insightful levels of abstract thinking and originality is shown in the implementation of a suitable model.		X2
2	Range of conflicting requirements: Involve wide-ranging or conflicting technical, engineering, and other issues.	Implements a solution tackling very basic problems and issues that have limited or no impact.	Implements a solution which tackles a few problems of conflicting nature but fails to address the	Implements a solution which tackles some wide-ranging issues and conflicting problems.	Implements a solution which tackles wide-ranging issues and conflicting problems.		X3

			other wide-ranging issues.				
4	Familiarity of issues: Involve infrequently encountered issues.	Demonstrates limited ability to involve infrequently encountered issues, the solution only addresses common basic issues	Demonstrates the ability to involve a few infrequently encountered issues but is unable to extend beyond some select issues.	Demonstrates an adequate ability to involve infrequently encountered issues, the presented solution addresses some rare issues.	Demonstrates the solution in an excellent manner show-casing the ability to involve and overcome the infrequently encountered issues.		X4
5	Extent of applicable codes: Are outside problems encompassed by standards and codes of practice?	The implemented solution lies within the problems laid out by the standards and codes of practice and is unable to show any dissimilarity from already encompassed problems.	Implemented solution shows little variance from already encompassed problems but is unable to meet the required criteria i.e., it falls under the already encompassed problems.	Implements an adequate solution which lies outside the already enclosed solutions.	An exemplary solution which addresses a unique and complex problem which lies outside the standards and codes of practice.	NA	X5
6	Extent of stakeholder involvement and level of conflicting requirements: Involve diverse groups of stakeholders with varying needs.	The solution lacks to include a diverse group of stakeholders catering only to the needs of a select few.	The solution includes some groups but lacks diversity of stakeholders failing to address the varying needs.	The solution addresses the varying needs of different diverse groups of stakeholders in a satisfactory manner.	The solution takes an excellent approach to meet the needs of a diverse group of stakeholders, the solution takes into account a varying number of possibilities and conditions which can be faced by various groups.	NA	X6
7	Consequences: Have significant consequences to society, environment and industry encompassing SDGs.	The implemented solution fails to show any form of significant impact or importance.	Implemented solution shows some level of importance but is unable to make any significant changes.	The implemented solution shows a certain level of significance but can be further improved to make a higher level of impact.	An impressive and intuitive approach to the solution, highlighting its level of significance.	NA	X7
8	Inter-dependence: Are high level problems including many parts.	Demonstrates a simple solution failing to have multiple parts, the level of ability shown is basic and lacks innovation.	Demonstrates a solution with little complexity, the level of ability is inadequate and requires further innovation.	The demonstrated solution shows promise in regards to its level of complexity but requires further improvement to meet the criteria of inter-dependence.	The demonstrated solution encompasses many different engineering aspects showcasing the complexity and inter-dependence of the solution	NA	X8
9	Judgement: Require judgement in decision making	The presented solution is straightforward and involves no judgement in decision making	An inadequate level of thought is put behind the presented solution and requires little judgement when making a decision.	Presented solution requires a moderate level of judgment when making decisions.	Presented solution involves a significant level of judgement in decision making.	NA	X9

Total $(=\frac{X1+X2+X3+X4}{20}$ x $Weight\ of\ the\ CEP)$		

7. CEP Mapping to CLO and WP, And Course CLO to Bloom's Taxonomy, PLO, WK, and SDG

Course Name	Course Code	CLO	WP	Domain / Taxonomy Level	PLO	WK	SDG
Fundamental of Databases	CS2011	CLO- 2, CLO-3	WP1, WP2, WP3, WP4	Cognitive Level 3 (C3)	PLO-1, PLO-5	WK4, WK6	Goal 9

7.1. Course Learning Outcomes CLO's

S. No.	CLO	Domain	Taxonomy Level	
1	Explain the basic concepts and uses of databases with different applications/ Systems.	Cognitive	2	
2	Build code of Structured Query Language (SQL) for database definition and manipulation using any DBMS.	Cognitive	3	✓
3	Describe and apply different stages of database development using different data models	Cognitive	3	✓
4	Explain database programming techniques	Cognitive	2	

7.2. Program Learning Outcomes PLO's

PLO	Attributes	
1	Engineering Knowledge	✓
2	Problem Analysis	
3	Design/Development of Solutions	
4	Investigation	
5	Modern Tool Usage	✓
6	The Engineer and Society	
7	Environment and Sustainability	
8	Ethics	
9	Individual and Team Work	
10	Communication	
11	Project Management	

12	Lifelong Learning	

7.3. Knowledge and Attributes Profile

Code	Description	
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.	
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.	
WK3	A systematic, theory-based formulation of engineering fundamentals required in the relevant engineering discipline.	
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.	✓
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.	
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.	✓
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development (Represented by the 17 UN Sustainable Development Goals (UN-SDG)).	
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.	
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.	

7.4. United Nations Social Development Goals SDG’s

Goal No	Goal	
1	No poverty	
2	Zero hunger (No hunger)	
3	Good health and well-being	
4	Quality education	
5	Gender equality	
6	Clean water and sanitation	
7	Affordable and clean energy	
8	Decent work and economic growth	
9	Industry, Innovation and Infrastructure	✓
10	Reduced inequality	

11	Sustainable cities and communities	
12	Responsible consumption and production	
13	Climate action	
14	Life below water	
15	Life on land	
16	Peace, justice and strong institutions	
17	Partnership for the goals	